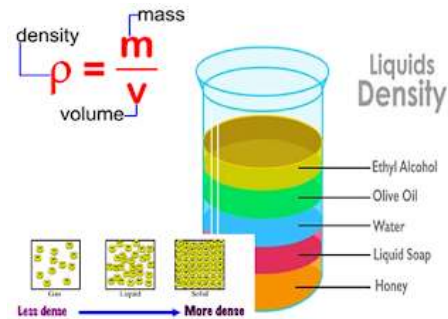




1. **Define density?** Ans: It is defined as the ratio of mass per unit volume of the fluid.
2. **Define viscosity?** Ans: It is defined as the property of fluid which offers resistance to the movement of fluid over another adjacent layer of the fluid.
3. **Differentiate between real fluids and ideal fluids?** Ans: A fluid, which is incompressible and is having no viscosity, is known as ideal fluid while the fluid, which possesses viscosity, is known as real fluid.
4. **What is a Venturimeter?** Ans: It is a device which is used for measuring the rate of flow of fluid flowing through pipe.
5. **What is a notch?** Ans: A notch is a device used for measuring the rate of flow of a fluid through a small channel or a tank.
6. **Define buoyancy?** Ans: When a body is immersed in a fluid, an upward force is exerted by the fluid on the body. This upward force is equal to the weight of the fluid displaced by the body.
7. **Define meta-centre?** Ans: It is defined as the point about which a body starts oscillating when the body is tilted by a small angle.
8. **Define a pump?** Ans: The hydraulic machine which converts the mechanical energy into hydraulic energy is called a pump.
9. **Define centrifugal pump?** Ans: The pump which converts the mechanical energy in to hydraulic energy, by means of centrifugal force acting on the fluid is known as centrifugal pump.
10. **Define reciprocating pump?** Ans: The pump which converts the mechanical energy in to hydraulic energy by sucking the liquid into a cylinder in which a piston is reciprocating, which exerts the thrust on the liquid and increases its hydraulic energy is known as reciprocating pump.
11. **What is impact of jet means?** Ans: It means the force exerted by the jet on a plate which may be stationary or moving.
12. **What is a turbine?** Ans: A turbine is a hydraulic machine which converts hydraulic energy in to mechanical energy.
13. **What is tangential flow turbine?** Ans: If the water flows along the tangent of the runner, the turbine is known as tangential flow turbine.
14. **What is radial flow turbine?** Ans: If the water flows in the radial direction through the runner, the turbine is called radial flow turbine.
15. **State Newton's law of viscosity?** Ans: It states that the shear stress on a fluid element layer is directly proportional to the rate of shear strain.
16. **What are the devices used for pressure measurement?** Ans: The devices used are manometers, diaphragm pressure gauge, dead weight pressure gauge etc.
17. **Why blower is used?** Ans: Blower is used to discharge higher volume of air at low pressure.
18. **State continuity equation?** Ans: For a fluid flowing through a pipe at all the cross section, the quantity of fluid per second is constant.
19. **What are the methods of describing fluid motion?** Ans: The fluid motion is described by two methods. They are lagrangian method and eulerian method.
20. **Where the notches are used?** Ans: Notches are usually used in tanks or small channels.
21. **What is a weir?** Ans: Weir is a concrete structure placed in an open channel over which the flow occurs.
22. **What do you understand by the term major loss in pipes?** Ans: When a fluid is flowing through a pipe, some of the energy is lost due to friction; this is termed as major loss.
23. **What do you understand by the term minor loss in pipes?** Ans: When a fluid is flowing through a pipe, some of the energy is lost due to sudden expansion of pipe, sudden contraction, bend and pipe fitting; these are termed as minor loss.
24. **Define the term hydraulic gradient?** Ans: It is defined as the line which gives the sum of pressure head and datum head of a flowing fluid in a pipe with respect to some reference line.
25. **Define the term total energy line?** Ans: It is defined as the line which gives the sum of pressure head, datum head and kinetic head of a flowing fluid in a pipe with respect to some reference line.
26. **What is a draft tube?** Ans: It is a pipe of gradually increasing area which connects the outlet of the runner to the tail race. It is used for discharging water from the exit of the turbine to the tail race.
27. **Define co-efficient of velocity of jet?** Ans: It is defined as the ratio between the actual velocity of a jet of liquid at vena-contracta and the theoretical velocity of jet.
28. **Define co-efficient of contraction of orifice meter?** Ans: It is defined as the ratio of the area of the jet at vena-contracta to the area of the orifice.
29. **Define co-efficient of discharge of orifice meter?** Ans: It is defined as the ratio of the actual discharge from an orifice to the theoretical discharge from the orifice.





30. **What is vena-contracta?** Ans: It is a section at which the stream lines are straight and parallel to each other and perpendicular to the plane of the orifice.
31. **Why the Centrifugal Pump is called High Discharge pump?** Ans: Centrifugal pump is a kinetic device. The centrifugal pump uses the centrifugal force to push out the fluid. So the liquid entering the pump receives kinetic energy from the rotating impeller. The centrifugal action of the impeller accelerates the liquid to a high velocity, transferring mechanical (rotational) energy to the liquid. So it discharges the liquid in high rate.
It is given in the following formula: Centrifugal force $F = (M \cdot V^2) / R$. Where, M-Mass; V-Velocity; R-Radius
32. **How Cavitations can be eliminated by Pump?** Ans: Cavitations means bubbles are forming in the liquid. To avoid Cavitation, we have to increase the Pump size to one or Two Inch; to increase the pressure of the Suction Head, or Decrease the Pump Speed.
33. **Why Cavitations will occur in Centrifugal Pump and not in Displacement Pump?** Ans: The formation of cavities (or bubbles) is induced by flow separation, or non-uniform flow velocities, inside a pump casing. In centrifugal pumps the eye of the pump impeller is smaller than the flow area of pipe. This decrease in flow area of pump results in increase in flow rate. So pressure drop happened between pump suction and the vanes of the impeller. Here air bubbles or cavities are formed because of liquid vapour due to increase in temperature in impeller. This air bubbles are transmitted to pump which forms cavitations.
34. **Which Pump is more Efficient Centrifugal Pump or Reciprocating Pump?** Ans: Centrifugal pump because flow rate is higher compared to reciprocating pump. Flow is smooth and it requires less space to install. Lower initial cost and lower maintenance cost.
35. **Why Centrifugal Pump is not called as a Positive Displacement Type of Pump?** Ans: The centrifugal has varying flow depending on pressure or head, whereas the Positive Displacement pump has more or less constant flow regardless of pressure. Likewise viscosity is constant for positive displacement pump where centrifugal pump have up and down value because the higher viscosity liquids fill the clearances of the pump causing a higher volumetric efficiency. When there is a viscosity change in supply there is also greater loss in the system. This means change in pump flow affected by the pressure change. One more example is, positive displacement pump has more or less constant efficiency, where centrifugal pump has varying efficiency rate.
36. **How Cavitations can be eliminated in a Pump?** Ans: Cavitations means bubbles are forming in the liquid. To avoid Cavitations, we have to increase the Pump size to one or Two Inch; to increase the pressure of the Suction Head, or Decrease the Pump Speed.
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39. **What is a radial-flow turbine?** Ans: In a radial-flow turbine, steam flows outward from the shaft to the casing. The unit is usually a reaction unit, having both fixed and moving blades.
40. **What are four types of turbine seals?** Ans: Carbon rings fitted in segments around the shaft and held together by garter or retainer springs. Labyrinth mated with shaft serration's or shaft seal strips. Water seals where a shaft runner acts as a pump to create a ring of water around the shaft. Use only treated water to avoid shaft pitting and stuffing box using woven or soft packing rings that are compressed with a gland to prevent leakage along the shaft.
41. **What are two types of clearance in a turbine?** Ans: Radial – clearance at the tips of the rotor and casing. Axial – the fore-and-aft clearance, at the sides of the rotor and the casing.
42. **What is the function of a thrust bearing?** Ans: Thrust bearings keep the rotor in its correct axial position.
43. **What is a stage in a steam turbine?** Ans: In an impulse turbine, the stage is a set of moving blades behind the nozzle. In a reaction turbine, each row of blades is called a "stage." A single Curtis stage may consist of two or more rows of moving blades.
44. **What is a diaphragm?** Ans: Partitions between pressure stages in a turbine's casing are called diaphragms. They hold the vane-shaped nozzles and seals between the stages. Usually labyrinth-type seals are used. One-half of the diaphragm is fitted into the top of the casing, the other half into the bottom.
45. **What are the two basic types of steam turbines?** Ans: Impulse type and Reaction type.



46. **What are topping and superposed turbines?** Ans: Topping and superposed turbines are high-pressure, non-condensing units that can be added to an older, moderate-pressure plant. Topping turbines receive high-pressure steam from new high-pressure boilers. The exhaust steam of the new turbine has the same pressure as the old boilers and is used to supply the old turbines.
47. **What is a combination thrust and radial bearing?** Ans: This unit has the ends of the Babbitt bearing extended radially over the end of the shell. Collars on the rotor face these thrust pads, and the journal is supported in the bearing between the thrust collars.
48. **How Cavitation can be eliminated in a Pump?** Ans: Increase the Pump size to one or Two Inch; Increase the pressure of the Suction Head; Decrease the Pump Speed.
49. **One litre = _____cm³ (1000 cm³)**
50. **Which Pump is more Efficient Centrifugal Pump or Reciprocating Pump?** Ans: Centrifugal pump because flow rate is higher compared to reciprocating pump. Flow is smooth and it requires less space to install. Lower initial cost and lower maintenance cost.
51. **When to use Centrifugal Pumps and Reciprocating Pumps?** Ans: **Centrifugal Pumps:** Best suited for applications requiring high flow and relatively low pressure, such as general water supply systems, irrigation, or circulation.
Reciprocating Pumps: Ideal for low-flow and extremely high-pressure applications, like dosing chemicals, hydraulic presses, or high-pressure cleaning.
52. **What is difference between centrifugal pumps and reciprocating pumps?** Ans: The primary difference is their working principle: **centrifugal pumps** use kinetic energy from a rotating impeller to continuously move fluids, while **reciprocating pumps** use back-and-forth piston or plunger movements to trap and physically displace a fixed volume of fluid.

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